

10. Ans: C

Escape speed, v

$$v = \sqrt{\frac{2GM}{r}}$$

$$v \propto \sqrt{\frac{M}{r}}$$

$$\frac{v_m}{v_E} = \sqrt{\frac{M_m}{M_E} \times \frac{r_E}{r_m}} = \sqrt{\frac{1}{81} \times \frac{4}{1}} = \frac{2}{9} = 0.22$$

11. Ans: A